

8 (a) Outline the purpose of lexical analysis during the compilation of a program.

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..... [2]

(b) Write the Reverse Polish Notation (RPN) for the given infix expression:

$$(2 - 6) * (13 + 7) / 5$$

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..... [2]

(c) The RPN expression:

$$d \ a \ b \ + \ * \ c \ a \ - \ /$$

is to be evaluated, where:

$$a = 6, \ b = 12, \ c = 15 \ \text{and} \ d = 5.$$

Show the changing contents of the stack as the RPN expression is evaluated.

[4]

